

A

Project Report

Submitted in partial fulfilment of the requirements for the completion of the Course

EDS Lab Codetantra

Practical No.4

(2304102L)

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4.1.1. Pandas - series creation and manipulation

03:12

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd
```

```
# # Take inputs from the user to create a list of numbers
```

```
# numbers = list(map(int, input().split()))
```

```
# grouped =
```

```
# # Display the mean of even and odd numbers with labels #
```

```
grouped.index = ['Even' if is_even else 'Odd' for is_even in
```

```
grouped.index] # print("Mean of even and odd numbers:")
```

```
# print(grouped) import
```

```
pandas as pd
```

```
# Take inputs from the user to create a list of numbers numbers
```

```
= list(map(int, input().split()))
```

```
# Create a Pandas series from the list of numbers
```

```
s = pd.Series(numbers)
```

```
# Grouping by even and odd numbers and calculating the mean grouped
```

```
= s.groupby(s % 2 == 0).mean()
```

```
# Display the mean of even and odd numbers with labels
```

```
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```
print("Mean of even and odd numbers:")
```

```
print(grouped)
```

The screenshot displays a code editor interface with a sidebar on the left labeled 'Explorer' and a sidebar on the right labeled 'Debugger'. The main area shows the test results for a Python program. At the top, there are two summary boxes: 'Average time' (0.007 s, 7.00 ms) and 'Maximum time' (0.017 s, 17.00 ms). Below these, there are two green checkmarks indicating '3 out of 3 shown test case(s) passed' and '3 out of 3 hidden test case(s) passed'. The first test case is expanded, showing 'Test case 1' with a duration of 17 ms. It includes a 'Debug' button and a 'Test' button. The 'Expected output' and 'Actual output' sections both show the same results: a list of 10 integers (1 2 3 4 5 6 7 8 9 10), the mean of even and odd numbers (5.0 and 6.0 respectively), and the dtype (float64). The second and third test cases are also shown as passed with durations of 5 ms each.

4.1.2. Dictionary to dataframe

07:26

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

Convert the dictionary to a Pandas DataFrame.

Add a new row:

Take inputs from the user for the new row data (name, age).

Add the new row to the DataFrame.

Display the DataFrame after adding the new row.

Modify a row:

Modify a specific row by changing the age. Take the row index and new age value from the user.

Display the DataFrame after modifying the row.

Delete a row:

Take the row index to be deleted from the user.

Remove the specified row.

Display the DataFrame after deleting the row.

Add a new column:

Add a column Gender with values taken from the user.

Display the DataFrame after adding the new column.

Modify a column:

Convert names to uppercase.

Display the DataFrame after modifying the column.

Delete a column:

Remove the Age column.

Display the DataFrame after deleting the column.

```
import pandas as pd
# Provided dictionary of lists data
= {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
```

```
}
```

```
# Convert the dictionary to a
```

```
DataFrame df = pd.DataFrame(data)
```

```
# Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Adding a new row new_name =
```

```
input("New name: ") new_age =
```

```
int(input("New age: ")) df.loc[len(df)] =
```

```
[new_name, new_age]
```

```
# Display the DataFrame after adding a new row print("After  
adding a row:\n",df)
```

```
# Modifying a row row_to_modify = int(input("Index  
of row to modify: ")) new_age_value = int(input("New  
age: ")) df.at[row_to_modify, "Age"] =  
new_age_value
```

```
# Display the DataFrame after modifying a row print("After  
modifying a row:")
```

```
print(df)
```

```
# Deleting a row row_to_delete = int(input("Index of  
row to delete: "))  
df = df.drop(index=row_to_delete).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row print("After  
deleting a row:")  
print(df)
```

```
# Adding a new column genders_input = input("Enter  
genders separated by space: ") genders =  
genders_input.split()
```

```
# Ensure correct length if  
len(genders) == len(df):  
    df["Gender"] = genders  
else:  
    print("Error: Number of genders must match rows!") df["Gender"]  
    = ["Unknown"] * len(df)
```

```
# Display the DataFrame after adding a new column  
print("After adding a new column:")
```

```
print(df)
```

```
# Modifying a column
```

```
df["Name"] = df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column print("After  
modifying a column:")
```

```
print(df)
```

```
# Deleting a column
```

```
df = df.drop(columns=["Age"])
```

```
# Display the DataFrame after deleting a column print("After  
deleting a column:")
```

```
print(df)
```


Explorer

Average time
0.043 s
43.00 ms

Maximum time
0.049 s
49.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 49 ms Debug

Expected output

Actual output

Original DataFrame:
.....Name...Age
0...Alice...25
1...Bob...30
2...Charlie...35
New name: Susan
New age: 29
After adding a row:
.....Name...Age
0...Alice...25
1...Bob...30
2...Charlie...35
3...Susan...29
Index of row to modify: 0
New age: 27
After modifying a row:
.....Name...Age
0...Alice...27
1...Bob...30
2...Charlie...35
3...Susan...29
Index of row to delete: 3
After deleting a row:
.....Name...Age
0...Alice...27
1...Bob...30

Original DataFrame:
.....Name...Age
0...Alice...25
1...Bob...30
2...Charlie...35
New name: Susan
New age: 29
After adding a row:
.....Name...Age
0...Alice...25
1...Bob...30
2...Charlie...35
3...Susan...29
Index of row to modify: 0
New age: 27
After modifying a row:
.....Name...Age
0...Alice...27
1...Bob...30
2...Charlie...35
3...Susan...29
Index of row to delete: 3
After deleting a row:
.....Name...Age
0...Alice...27
1...Bob...30

Terminal

Test cases

Average time: 0.043 s
Maximum time: 0.049 s

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

After modifying a row:

	Name	Age
0	Alice	27
1	Bob	30
2	Charlie	35
3	Susan	29

Index of row to delete: 3

After deleting a row:

	Name	Age
0	Alice	27
1	Bob	30
2	Charlie	35

Enter genders separated by space: F M M

After adding a new column:

	Name	Age	Gender
0	Alice	27	F
1	Bob	30	M
2	Charlie	35	M

After modifying a column:

	Name	Age	Gender
0	ALICE	27	F
1	BOB	30	M
2	CHARLIE	35	M

After deleting a column:

	Name	Gender
0	ALICE	F
1	BOB	M
2	CHARLIE	M

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd

# Read the text file into a DataFrame
file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age",
"Grade"]) # Display the first five rows
print("First five rows:")
print(data.head())

# Calculate the average age (2 decimal places)
avg_age = round(data["Age"].mean(), 2)
print("Average age:", avg_age)

# Filter students with grade up to 'B' (A and B) filtered
= data[data["Grade"].isin(['A', 'B'])]
```

```
print("Students with a grade up to B")
```

```
print(filtered)
```

Average time: 0.028 s, 27.50 ms | Maximum time: 0.040 s, 40.00 ms | 1 out of 1 shown test case(s) passed | 1 out of 1 hidden test case(s) passed

Test case 1 (40 ms) [Debug] [Test cases]

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
...	...
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
3 Mike 23 C	3 Mike 23 C
4 Sony 21 B	4 Sony 21 B
Average age: 23.29	Average age: 23.29
Students with a grade up to B	Students with a grade up to B
...	...
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
4 Sony 21 B	4 Sony 21 B
5 Amia 26 A	5 Amia 26 A

Terminal | Test cases | < Prev | Reset | Submit | Next >

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd

# Prompt the user for the file name file_name
file_name = input()

# Load the data df =
df = pd.read_csv(file_name) # Prepare
data for calculation df['Date'] =
pd.to_datetime(df['Date']) df['Month']
= df['Date'].dt.to_period('M')
df['Sales'] = df['Quantity'] * df['Price']

# Find the month with the highest total sales best_month
best_month = df.groupby('Month')['Sales'].sum().idxmax()
highest_sales = df.groupby('Month')['Sales'].sum().max()
```

```
print(f"Best month: {best_month}") print(f"Total
sales: ${highest_sales:.2f}")
```



4.2.2. Best Selling

Product 02:08

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Find the product that sold the most in terms of quantity sold.

Display the product that sold the most and the total quantity sold for that product.

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New

York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New

York 2025-01-02,Product

C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product **B**,5,15,Los

Angeles

2025-01-05,Product **A**,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the data df =
```

```
pd.read_csv(file_name)
```

```
# Find the product with the highest total quantity sold
```

```
best_product =
```

```
df.groupby('Product')['Quantity'].sum().idxmax()
```

```
highest_quantity =
```

```
df.groupby('Product')['Quantity'].sum().max() # Display the
```

```
result print(f"Best selling product: {best_product}")
```

```
print(f"Total quantity sold: {highest_quantity}")
```



4.2.3. City that Sold the Most

Products 01:45

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by City and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New

York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New

York 2025-01-02,Product

C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los

Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the data df =
```

```
pd.read_csv(file_name)
```

```
# Group by City and calculate total quantity city_sales
```

```
= df.groupby('City')['Quantity'].sum()
```

```
# Find the city with the highest total quantity sold best_city
```

```
= city_sales.idxmax()
```

```
# Display the result print(f"City sold the most
```

```
products: {best_city}")
```



4.2.4. Most Frequently Sold Product

Pairs 01:25

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pair/s that was sold most frequently.

Sample Data :

Date,Product,Quantity,Price,City

2025-01-01,Product **A**,5,20,New

York

2025-01-01,Product **B**,3,15,Los Angeles

2025-01-02,Product **A**,7,20,New

York 2025-01-02,Product

C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product **A**,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product **B**,5,15,Los

Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Explanation:

Transactions:

2025-01-01: Product A, Product B

2025-01-02: Product A, Product C

2025-01-03: Product B, Product A

2025-01-04: Product C, Product B

2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

Product A and Product B : Appear in transactions on 2025-01-01 and 2025-01-03.

Product A and Product C : Appear in transactions on 2025-01-02 and 2025-01-05.

Product B and Product C : Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

Product A and Product B (2 times)

Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd

from itertools import combinations from
collections import Counter

# Prompt user to input the file name file_name
= input()

# Read data from the specified CSV file
df = pd.read_csv(file_name) # write the
code pair_counter = Counter()

# Group by Date and generate product pairs
for _, group in df.groupby('Date'): products =
group['Product'].tolist() pairs =
combinations(sorted(products),
    2) pair_counter.update(pairs)

# Find maximum frequency
max_count = max(pair_counter.values())

# Output the most frequent product pairs
for pair, count in pair_counter.items(): if
count == max_count:
```

```
print(f"{pair[0]} and {pair[1]}: {count} times")
```

Average time

0.020 s

20.33 ms

Maximum time

0.039 s

39.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 **39 ms** **Debug**

Expected output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Actual output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Terminal

Test cases

4.2.5. Titanic Dataset Analysis and Data

Cleaning 01:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
```

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset
```

```
data =
```

```
pd.read_csv('Titanic-Dataset.csv') # 1.
```

```
Display the first 5 rows of the dataset
```

```
print(data.head())
```

```
# 2. Display the last 5 rows of the
```

```
dataset print(data.tail())
```

```
# 3. Get the shape of the dataset print(data.shape)
```

```
# 4. Get a summary of the dataset (info)
```

```
print(data.info())
```

5. Get basic statistics of the dataset `print(data.describe())`

6. Check for missing values

`print(data.isnull().sum())`

7. Fill missing values in the 'Age' column with the median age `data['Age'] = data['Age'].fillna(data['Age'].median())`


```
# 8. Fill missing values in the 'Embarked' column with the mode data['Embarked']
= data['Embarked'].fillna(data['Embarked'].mode()[0])
```

```
# 9. Drop the 'Cabin' column due to many missing values
data = data.drop(columns=['Cabin'])
```

```
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch' data['FamilySize']
= data['SibSp'] + data['Parch']
```

The screenshot shows a Jupyter Notebook interface with a sidebar on the left labeled 'Explorer'. The main area displays a test case result: 'Test case 1' with a green checkmark, a duration of '166 ms', and a 'Debug' button. Below this, a table compares 'Expected output' and 'Actual output' for a data transformation. The table has 12 columns: PassengerId, Survived, Pclass, Fare, Cabin, Embarked, and FamilySize. The 'Expected output' table shows 5 rows of data, and the 'Actual output' table shows 5 rows of data. The data is identical in both tables, indicating the transformation was successful. The 'FamilySize' column is calculated as the sum of 'SibSp' and 'Parch'.

PassengerId	Survived	Pclass	Fare	Cabin	Embarked	FamilySize
0	1	0	3	7.2500	NaN	S
1	2	1	1	71.2833	C85	C
2	3	1	3	7.9250	NaN	S
3	4	1	1	53.1000	C123	S
4	5	0	3	8.0500	NaN	S

[5 rows x 12 columns]

PassengerId	Survived	Pclass	Fare	Cabin	Embarked	FamilySize
886	887	0	2	13.00	NaN	S
887	888	1	1	30.00	B42	S
888	889	0	3	23.45	NaN	S
889	890	1	1	30.00	C148	C
890	891	0	3	7.75	NaN	Q

[5 rows x 12 columns]

Explorer

Average time
0.166 s
166.00 ms

Maximum time
0.166 s
166.00 ms

1 out of 1 shown test case(s) passed

Debugger

[5 rows x 12 columns]	[5 rows x 12 columns]
(891, 12)	(891, 12)
<class 'pandas.core.frame.DataFrame'>	<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890	RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):	Data columns (total 12 columns):
# Column Non-Null Count Dtype	# Column Non-Null Count Dtype
0 PassengerId 891 non-null int64	0 PassengerId 891 non-null int64
1 Survived 891 non-null int64	1 Survived 891 non-null int64
2 Pclass 891 non-null int64	2 Pclass 891 non-null int64
3 Name 891 non-null object	3 Name 891 non-null object
4 Sex 891 non-null object	4 Sex 891 non-null object
5 Age 714 non-null float64	5 Age 714 non-null float64
6 SibSp 891 non-null int64	6 SibSp 891 non-null int64
7 Parch 891 non-null int64	7 Parch 891 non-null int64
8 Ticket 891 non-null object	8 Ticket 891 non-null object
9 Fare 891 non-null float64	9 Fare 891 non-null float64
10 Cabin 204 non-null object	10 Cabin 204 non-null object
11 Embarked 889 non-null object	11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)	dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB	memory usage: 66.2+ KB
None	None
PassengerId Survived Pclass SibSp Parch Fare	PassengerId Survived Pclass SibSp Parch Fare
count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000	count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000
mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208	mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208
std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429	std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429

Terminal

Test cases

The screenshot shows a Jupyter Notebook interface. At the top, a status bar indicates '1 out of 1 shown test case(s) passed'. Below this, there are two panels showing statistical data for a dataset. The left panel shows the following statistics: count (891.000000), mean (446.000000), std (257.353842), min (1.000000), 25% (223.500000), 50% (446.000000), 75% (668.500000), and max (891.000000). The right panel shows the same statistics. Below the statistics, there is a data preview table with 8 rows and 7 columns. The columns are PassengerId, Survived, Pclass, Name, Sex, Age, and SibSp. The data is as follows:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
0	0	0	0	0	177	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

At the bottom of the notebook, there is a terminal and a test cases section. The terminal is empty, and the test cases section shows a single test case that passed. The bottom of the screen shows a Windows taskbar with the time 9:37 PM and date 4/24/2026.

4.2.6. Titanic Dataset Analysis and Data Cleaning -

2 02:18

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.

7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd import
```

```
numpy as np
```

```
# Load the Titanic dataset data = pd.read_csv('Titanic-  
Dataset.csv') data['FamilySize'] = data['SibSp'] +  
data['Parch'] # 1. Create a new column 'IsAlone' (1 if alone,  
0 otherwise) data['IsAlone'] = np.where(data['FamilySize']  
== 0, 1, 0)
```

```
# 2. Convert 'Sex' to numeric (male: 0, female: 1) data['Sex']  
= data['Sex'].map({'male': 0, 'female': 1})
```

```
# 3. One-hot encode the 'Embarked' column  
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# 4. Get the mean age of passengers  
print(data['Age'].mean())
```

```
# 5. Get the median fare of passengers  
print(data['Fare'].median())
```

```
# 6. Get the number of passengers by class print(data['Pclass'].value_counts())
```

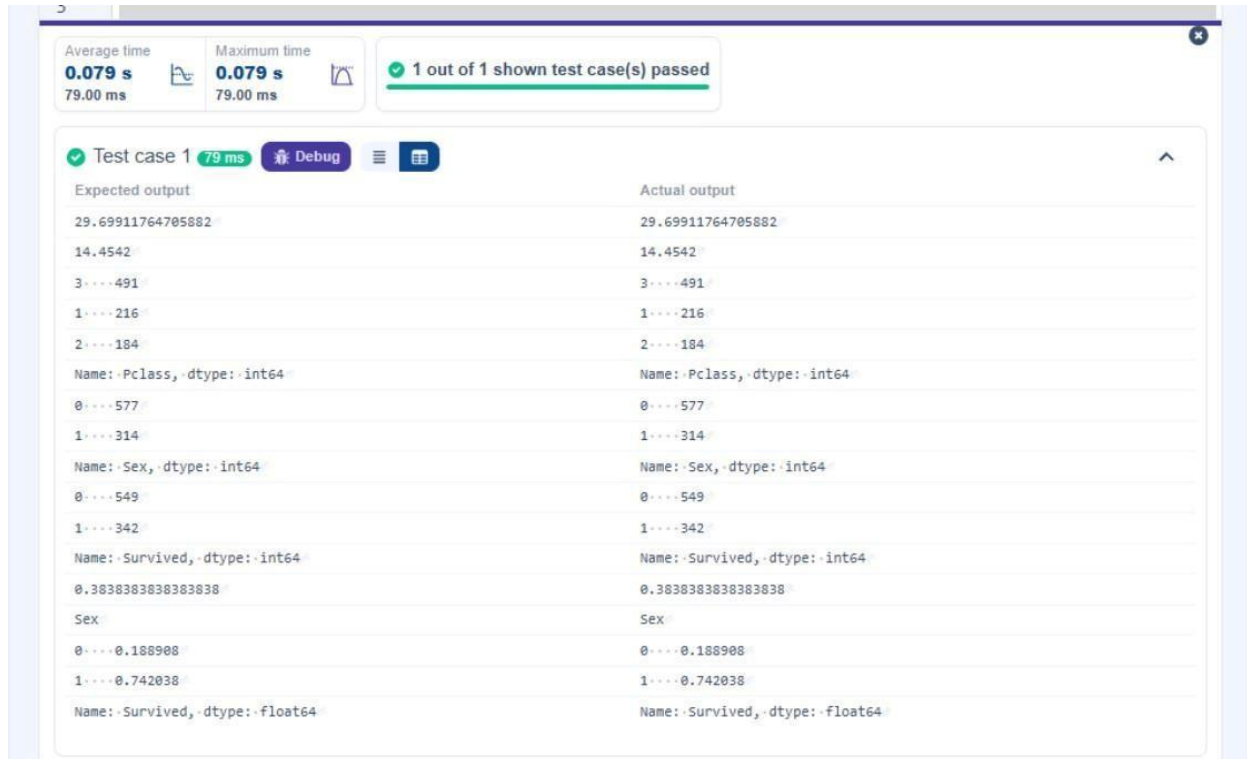
```
# 7. Get the number of passengers by gender print(data['Sex'].value_counts())
```

```
# 8. Get the number of passengers by survival status  
print(data['Survived'].value_counts())
```

9. Calculate the survival rate

```
print(data['Survived'].mean())
```

10. Calculate the survival rate by gender `print(data.groupby('Sex')['Survived'].mean())`



4.2.7. Titanic Dataset Analysis and Data Cleaning -

3 03:10

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).

8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset data = pd.read_csv('Titanic-
Dataset.csv') data['FamilySize'] = data['SibSp'] +
data['Parch'] data['IsAlone'] =
np.where(data['FamilySize'] > 0, 0, 1) data =
pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # 1. Calculate the survival rate by
class print(data.groupby('Pclass')['Survived'].mean())
```


2. Calculate the survival rate by embarked location

```
print(data.groupby('Embarked_S')['Survived'].mean())
```

3. Calculate the survival rate by family size

```
print(data.groupby('FamilySize')['Survived'].mean())
```

4. Calculate the survival rate by being alone

```
print(data.groupby('IsAlone')['Survived'].mean())
```

5. Get the average fare by class

```
print(data.groupby('Pclass')['Fare'].mean())
```

6. Get the average age by class `print(data.groupby('Pclass')['Age'].mean())`

7. Get the average age by survival status

`print(data.groupby('Survived')['Age'].mean())` # 8. Get the average fare by survival

status `print(data.groupby('Survived')['Fare'].mean())`

9. Get the number of survivors by class `print(data[data['Survived']`

`==`

`1]['Pclass'].value_counts())`

10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Explorer

Debugger

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

Expected output

Actual output

Pclass	Pclass
1...0.629630	1...0.629630
2...0.472826	2...0.472826
3...0.242363	3...0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0...0.506073	0...0.506073
1...0.336957	1...0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0...0.303538	0...0.303538
1...0.552795	1...0.552795
2...0.578431	2...0.578431
3...0.724138	3...0.724138
4...0.200000	4...0.200000
5...0.136364	5...0.136364
6...0.333333	6...0.333333
7...0.000000	7...0.000000
10...0.000000	10...0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0...0.505650	0...0.505650
1...0.303538	1...0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1...84.154687	1...84.154687
2...20.662183	2...20.662183
3...13.675550	3...13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1...38.233441	1...38.233441
2...29.877630	2...29.877630

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

5	0.136364	5	0.136364
6	0.333333	6	0.333333
7	0.000000	7	0.000000
10	0.000000	10	0.000000
Name: Survived, dtype: float64		Name: Survived, dtype: float64	
IsAlone		IsAlone	
0	0.505650	0	0.505650
1	0.303538	1	0.303538
Name: Survived, dtype: float64		Name: Survived, dtype: float64	
Pclass		Pclass	
1	84.154687	1	84.154687
2	20.662183	2	20.662183
3	13.675550	3	13.675550
Name: Fare, dtype: float64		Name: Fare, dtype: float64	
Pclass		Pclass	
1	38.233441	1	38.233441
2	29.877630	2	29.877630
3	25.140620	3	25.140620
Name: Age, dtype: float64		Name: Age, dtype: float64	
Survived		Survived	
0	30.626179	0	30.626179
1	28.343690	1	28.343690
Name: Age, dtype: float64		Name: Age, dtype: float64	
Survived		Survived	
0	22.117887	0	22.117887
1	48.395408	1	48.395408
Name: Fare, dtype: float64		Name: Fare, dtype: float64	
1	136	1	136
3	119	3	119
2	87	2	87
Name: Pclass, dtype: int64		Name: Pclass, dtype: int64	
3	372	3	372
2	97	2	97
1	80	1	80